



American International University-Bangladesh (AIUB)
Department of Computer Science
Faculty of Science & Technology (FST)

**Smart Integrated Voting System with Voter Information
Management & Public Opinion Wall**

A Software Engineering Project Submitted
By

Semester: Summer 24-25		Section:	Group Number:	
SL	Student Name	Student ID	Contribution (CO3+CO4)	Individual Marks
1	SUSMITA MITRA	23-50599-1		
2	ARNOB DEBNATH	23-50623-1		
3	SUDIPTO KUMAR CHAKRABARTY	23-50622-1		
4				
5				

The project will be evaluated for the following Course Outcomes

CO3 (PO-g-1) <i>Select appropriate software engineering models, project management roles and their associated skills for the complex software engineering project and evaluate the sustainability of developed software, taking into consideration the societal and environmental aspects</i>	Total Marks	
Selection of Software Engineering Models: Process model selection and presents sufficient evidence to support argument for the model selection	[5 Marks]	
Role identification and Responsibility Allocation: Well-planned project with proper role identification and responsibility allocation in the project management activities	[5Marks]	
Formatting and Submission: Submission, Defense, Completeness, Spelling, grammar, and Organization of the Project report	[5Marks]	
CO4 (PO-k-1) <i>Apply engineering management principles and economic decision making to develop software engineering project management plan.</i>	Total Marks	

Project WBS and Testcases: Relevant WBS (project task list) and testcases for the proposed project are stated properly.	[5Marks]	
Effort Estimation and Scheduling: Project estimation was described using proper effort estimation or schedules based on available project resources	[5Marks]	
Risk Management: Sufficient and appropriate risks are identified, analyzed, and properly categorized or prioritized.	[5Marks]	

1. PROJECT PROPOSAL

Smart Integrated Voting System with Voter Information Management & Public Opinion Wall

1.1 Background to the Problem

In a democratic country, every citizen has the important right to vote freely, fairly, and easily. But many countries, especially those with large populations, still use the old way of voting, where voters have to go to polling stations in person on a specific day. The main problem is that millions of people cannot vote due to various reasons, such as:

- Polling stations are far away in villages or remote areas
- Difficulty traveling because of physical disability or old age
- Safety issues in some areas
- Busy work or family responsibilities that make it hard to stand in long lines
- Natural disasters or situations like pandemics that limit movement

As a result, voter participation drops, which weakens democracy and raises questions about the fairness of the election. Nowadays, digital technology is widely used, so it is crucial to create a secure online voting system while maintaining offline voting options for those who cannot participate online. The problem now is that there is no trusted and secure digital voting system that is directly connected to the national Election Commission. Easy for everyone to use, protects voter identity, prevents fraud, and keeps the whole process transparent. On the other hand, though our election commission system is everything online, people still cannot change their valuable information directly online; the process is very complex and traditional. Even in the age of digital technology, people still have to go and submit the necessary documents in person. So it is very painful for the people. People faces many problems such as

- Travel from one place to another to submit documents.
- Time-consuming matter.

- Incorrect submission of documents results in having to reapply and submit documents, which is very painful for the people.

In this system, people can easily change their information, excluding specific information like their ID number or other information. Then the control manager can review their changing information. So this process is fully online and secure.

On the other hand, sometimes the government makes biased decisions, so the government needs to know what the people want, whether they agree with the decision or not. Therefore, a public opinion wall is very important where the government gathers feedback from the people, and people can express their opinions on specific questions or even vote on them. Then, based on what the people say, the government can better understand what the people want.

1.2 Solution to the Problem and Process Model Selection

Project Objective and Proposed Solution: The primary objective of this project is to develop a secure, transparent, and user-friendly digital voting and voter information management system that is directly integrated with the national Election Commission. The system will enable:

- Citizens to vote online securely and remotely
- Citizens to request updates to their voter information online
- Authorities to verify and approve changes securely
- Government to collect public feedback through a digital opinion platform

This solution aims to ensure increased participation in the democratic process, reduce logistical barriers, and enhance the transparency and responsiveness of government decisions.

This Solution Is Appropriate and Feasible: The proposed solution is appropriate because it directly addresses the key barriers faced by voters in the current traditional system:

- **Accessibility:** Voters from rural areas, people with disabilities, and the elderly can participate without needing to travel.
- **Convenience:** Citizens can request information updates and vote without standing in long queues or taking leave from work.
- **Security:** The system will include multi-factor authentication techniques.
- **Public opinion:** A dedicated opinion wall will enhance transparency in policymaking.

This solution is possible with current technology and infrastructure. The government already maintains a digital Election Commission system; this project proposes extending it with modern features that align with business goals like inclusivity, efficiency, and national trust.

Core Functionalities Using Modern Technology

1. Secure Online Voting System

- Biometric login, OTP, and voter ID verification
- Encrypted vote casting and counting
- Real-time participation tracking

2. Online Voter Information Update System

- Citizens can request updates to name, address, etc. (excluding ID)
- Backend review system for approval by control managers
- Document upload, validation, and automatic feedback

3. Public Opinion Platform

- Citizens can vote or comment on government proposals
- Analytics dashboard for government insight
- Anonymous but authentic participation

The system contributes to inclusive participation, public health (by enabling remote access), legal transparency, and cultural inclusiveness.

Target Users and Benefits:

Target Group

- Rural residents
- Elderly and physically challenged individuals
- Citizens with busy schedules
- Youth and first-time voters
- Government policymakers

Benefits

- Voting percentage increase, which indicates the inclusive participation that is very important for a democratic country
- Faster and more efficient voter information updates
- Increased public trust in elections and government decisions
- Inclusive and fair participation in policy discussions

Scientific Contribution: This project contributes to the field of e-governance, digital security, and public engagement by offering a practical, scalable solution.

- Demonstrates integration of voter authentication with digital platforms
- Promotes digital transparency in governmental feedback collection
- Bridges gaps between citizens and election administrators

Literature Review:

Blockchain for Electronic Voting System—Review and Open Research Challenges: This article highlights the growing interest in online voting due to its cost-effectiveness and accessibility, but also emphasizes the security risks involved. It introduces blockchain technology as a promising solution for secure, and verifiable electronic voting systems. While blockchain offers strong protection against fraud and manipulation, challenges like privacy and transaction speed still limit its scalability. The study concludes that current blockchain frameworks need improvement to be fully viable for national elections.

Compared to this proposal, both aim to solve voter accessibility and security issues, but this system goes further by integrating voter information updates and a public opinion wall for government feedback. A broader and more citizen-focused approach is presented in the proposal, while a more technical perspective centered on blockchain's role in voting security is offered in the article.

Existing Software Solutions and Their Limitations: Blockchain-based voting systems offer decentralized and verifiable election processes. They are used in little projects and small-scale elections.

Limitations:

- Complexity in implementation for large-scale national elections.
- Do not provide a public opinion feedback wall for democratic engagement

- Lack of integration with government databases

Extension and Improvement over Existing Solutions:

- Combines secure online voting, personal data update, and public opinion integrated within a system
- Offers a better user experience than existing systems
- Improves citizen and government trust and good communication

SOFTWARE DEVELOPMENT LIFE CYCLE

Process Model

For the development of the Smart Integrated Voting System with Voter Information Management & Public Opinion Wall, the Incremental Process Model has been selected.

Reason for Using the Incremental Model

- **Modular Development:**
The system is built in smaller, manageable parts (like voter information, online voting, opinion wall) so each can be handled individually.
- **Early Delivery:**
Basic features can be completed and shared early for testing and feedback before the full system is finished.
- **Flexibility to Changes:**
New or updated requirements can be added in future phases without affecting completed parts.
- **Risk Reduction:**
Problems can be found and fixed early in each part, preventing bigger issues later.
- **Efficient Use of Resources:**
Building one part at a time saves time and effort compared to building everything at once.
- **Improved Quality:**
Each part is properly tested before moving to the next phase, which makes the system more stable and reliable.
- **Suitable for Government Projects:**
Step-by-step progress, formal approval at each phase, and strong security needs are better supported by this model.

Reason for not using other models

- **Waterfall:** It follows a strict step-by-step process. Our project needs more testing part by part, which is not supported by the waterfall model.
- **Agile:** Agile needs continuous collaboration, rapid changes, and quick feedback. Government projects follow formal procedures, so Agile's fast-paced nature doesn't fit well.
- **Scrum:** Scrum involves daily meetings, sprints, and active stakeholder involvement. In a government setting, such frequent interaction and decision-making are not always possible.

1.3 Project Role Identification and Responsibilities

Our project group consists of three members — Susmita Mitra, Arnob Debnath, and Sudipto Kumar Chakrabarty. Each member played an important role in completing the Smart Integrated Voting System with Voter Information Management & Public Opinion Wall project. The work was divided equally so that everyone contributed to planning, development, testing, and documentation.

Sudipto Kumar Chakrabarty (System Analyst and Software Tester)

Sudipto worked as the System Analyst and Software Tester. At the beginning of the project, he collected and analyzed all system requirements. He was responsible for preparing the Software Requirement Specification (SRS), identifying both functional and non-functional requirements such as security, scalability, and performance. He helped define the system's core functionalities, including online voting, voter information updates, and the public opinion wall. Sudipto also helped select the Incremental Process Model, highlighting its benefits in modular development, flexibility, and reducing risks. After development, he prepared and executed all test cases to verify that each module worked correctly. He tested modules such as login, registration, information update, and opinion wall to ensure accuracy and reliability. Additionally, Sudipto applied software product metrics (Function Points, Maintainability Index, and Reliability metrics) to evaluate the overall software quality. He documented test results and contributed to the final review of the report.

Susmita Mitra (Project Manager and UI/UX Designer)

Susmita served as the Project Manager and UI/UX Designer. She took leadership of the entire project, managed communication between members, and maintained the timeline and task distribution. She led the discussion on model selection and finalized the Incremental Process Model with team approval. Susmita designed the user interface of the system, including pages for login, registration, online voting, information update, and the public opinion wall. She ensured the interface was clean, accessible, and user-friendly for both desktop and mobile users. She was also responsible for report organization, formatting, and final submission. She checked spelling, structure, and presentation to make sure the report met the university's guidelines. Additionally, Susmita managed the Git workflow, documentation, and coordinated the defense preparation, ensuring all project materials were complete and professional.

Arnob Debnath (Backend Developer and Database Engineer)

Arnob worked as the Backend Developer and Database Engineer. He was responsible for the system development and implementation part of the project. He designed and developed the backend logic for online voting, voter verification, and data management. Arnob created and managed the database structure, connecting modules like voter registration, election management, and public opinion storage. He implemented the security features of the system, including voter ID verification, OTP authentication, and data encryption, to ensure secure communication between client and server. Arnob also prepared the effort estimation and scheduling part using the COCOMO model, calculating total effort, duration, and team requirements. He participated in integrating different modules developed by the team and ensured smooth operation through database optimization. He also reviewed and tested the backend to make sure data was processed correctly and efficiently.

2. SOFTWARE REQUIREMENTS SPECIFICATIONS (SRS) / PRODUCT REQUIREMENTS DOCUMENT (PRD)

Citizen

- Online Registration with National ID verification.
- Login using OTP, biometric, with 2-Factor Authentication.
- Vote in elections (allowed once per election).
- View election results only after the official announcement.
- Update personal information (like name spelling, address, phone, email) but not critical fields like National ID or Date of Birth.
- Participate in Public Opinion Wall polls (vote/comment on government questions).

Election Commission Admin

- Approve or reject voter registration requests.
- Manage voter database (add, update, delete) with full audit logging.
- Create and manage elections (add candidates, set timelines, define voting rules).
- Monitor live voting progress and detect fraud attempts.
- Approve or reject voter information update requests.

Government / Opinion Wall Admin

- Post questions or polls for public feedback.
- Monitor citizen responses in real time.
- Generate analytics reports (voting percentage, engagement, trending topics).

System Super Admin

- Manage all user roles (Election Commission + Government admins).
- Control system-wide security settings and database configurations.
- Oversee audit logs and respond to critical incidents.

2.1 Functional Requirements

Online Voting

- Secure login with voter ID + OTP/biometric verification.
- Each voter can cast only one vote per election.
- Votes are encrypted to ensure anonymity.
- Real-time vote count is hidden until the election officially ends.
- Polling stations use the same database.

Voter Information Management

- Citizens can submit online applications to update (Name spelling, Address, Phone number, Email)

- Upload supporting documents (PDF/JPEG).
- Requests are forwarded to the Control Manager/Admin for review and approval.
- Voters receive notifications on approval or rejection.

Public Opinion Wall

- Government posts opinion polls or policy-related questions.
- Citizens can vote by (Yes/No/Neutral and Comment).
- Real-time analytics dashboard shows vote percentages, most discussed topics, and public opinion percentages.

2.2 Non-Functional Requirements

Performance & Scalability

- Must support 200 million users nationwide.
- Handle 70–80 million concurrent users during peak voting hours.
- System response time: < 1.5 seconds.

Reliability & Availability

- Minimum 99.99% uptime.
- Distributed & backups server

Security Features

- Two-Factor Authentication (2FA) via OTP & biometric (fingerprint/face recognition).
- End-to-End Encryption for all data transmission.
- Automated fraud detection (Multiple login attempt tracking, IP/device fingerprinting to detect suspicious patterns)

Usability

- Multi-language interface (Bangla & English).
- Accessible for elderly and disabled users (voice assistance, large fonts, high contrast).

- Responsive design for mobile, tablet, and desktop.

Compliance

- Must meet Bangladesh Election Commission regulations.
- Follow the Digital Security Act and National Data Privacy Laws.

Maintainability

- Modular architecture for easy updates, bug fixes, and future expansion.
- Detailed documentation and audit-ready logs.

Disaster Recovery

- Automatic daily encrypted backups.
- Recovery Time Objective: 15 minutes.

3. PROJECT ESTIMATION AND SCHEDULING

3.1 Effort and Cost Estimation

COCOMO (Constructive Cost Model):

Software Project Type: Embedded

Our project is classified as embedded software because it directly controls hardware components (such as sensors, biometric devices, and voting terminals) and runs within a dedicated system. This ensures real-time operation, reliability, and security, which are essential for accurate voter authentication and vote recording.

Coefficient:3.6

Project Complexity (P): 1.20

SLOC-dependent Coefficient (T): 0.32

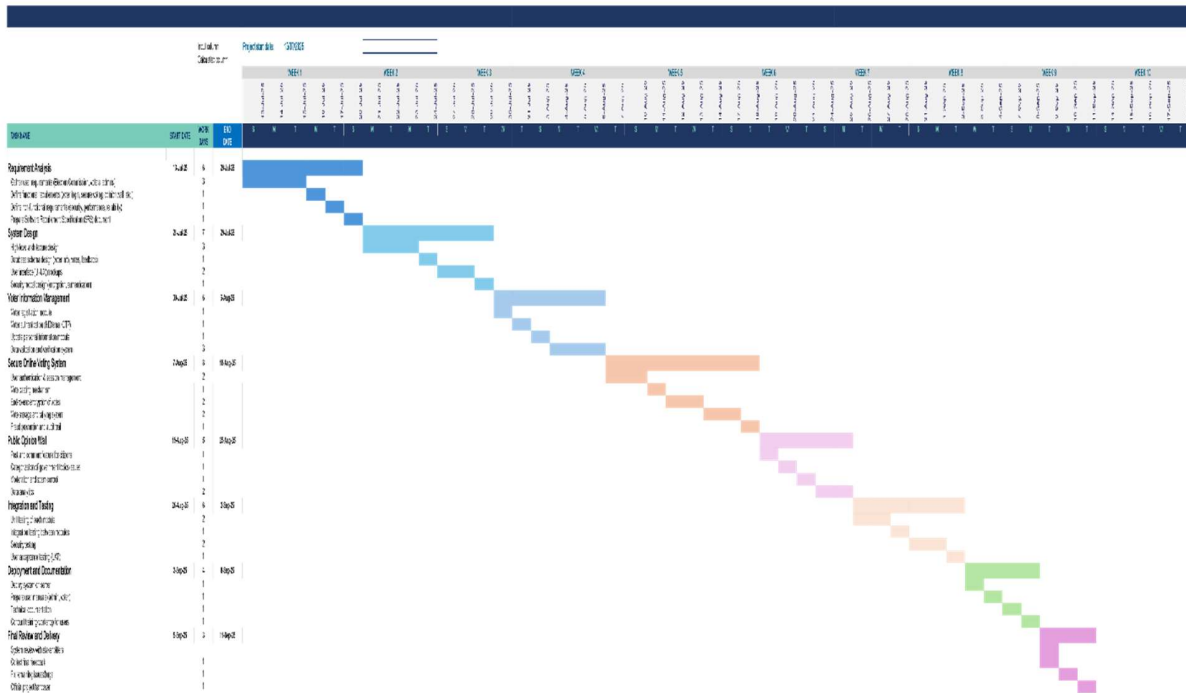
$$\begin{aligned} \text{Effort} = PM &= \text{Coefficient} * (\text{SLOC}/1000)^P & \text{SLOC} &= 10k \\ &= 3.6 * (10)^{1.20} \\ &= 57.05 \end{aligned}$$

$$\begin{aligned} \text{Development time} = DM &= 2.50 * (PM)^T \\ &= 2.50 * (57.05)^{0.32} \\ &= 9.11 \text{ week} \end{aligned}$$

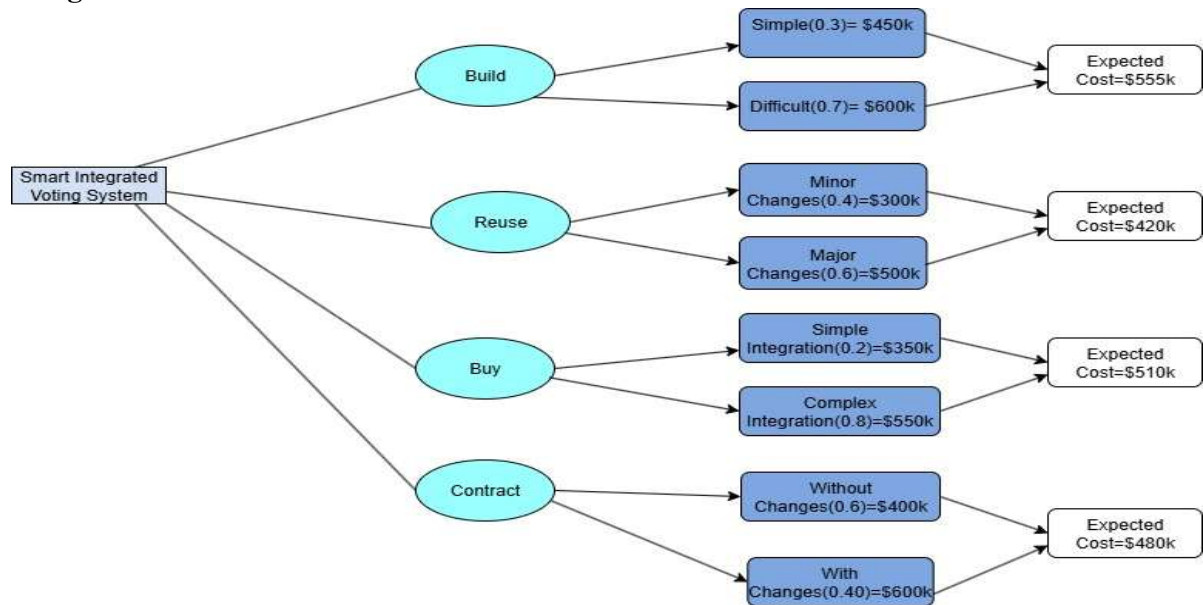
$$\begin{aligned} \text{Required number of people} = ST &= PM/DM \\ &= 57.05/9.11 \\ &= 6.26 \end{aligned}$$

3.2 Project Scheduling

Gantt Chart: <https://1drv.ms/x/c/acd313dac670c634/EWjXzMfPstAguHjtrr1DccB4K9gLghRYLtiTNS4dePgiw?e=DSfirt>



Budget:



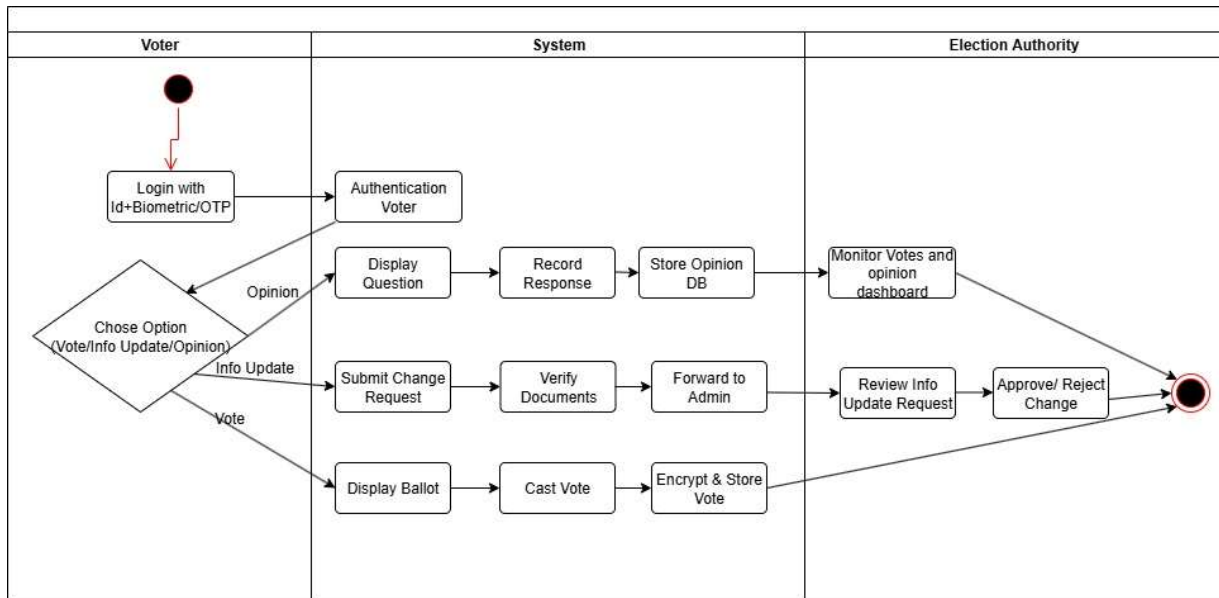
Risk:

Risk Name	Likelihood	Impact	Mitigation Plan
System Security Breach (Hacking, Fraud)	Medium	High	Use end-to-end encryption, multi-factor authentication, and regular testing
Server Downtime	Medium	High	Deploy load balancers, distributed servers, maintain backup servers, ensure 24/7 monitoring, and implement a disaster recovery plan.
Integration with the National Election Commission	Medium	High	Continuous communication with government IT teams, module integration.
Fake or Duplicate Voter Accounts	Low	High	Use biometric verification, OTP, and voter ID database cross-checking.
Low Internet Accessibility in Rural Areas	High	Medium	Enable offline voting, Center, mobile-friendly application, low bandwidth optimization.
Biased or Manipulated Public Opinion Wall	Medium	Medium	Transparency in Opinion
Project Delays due to Scope Changes	Low	Medium	Use an incremental model, prioritize features, and regularly progress tracking

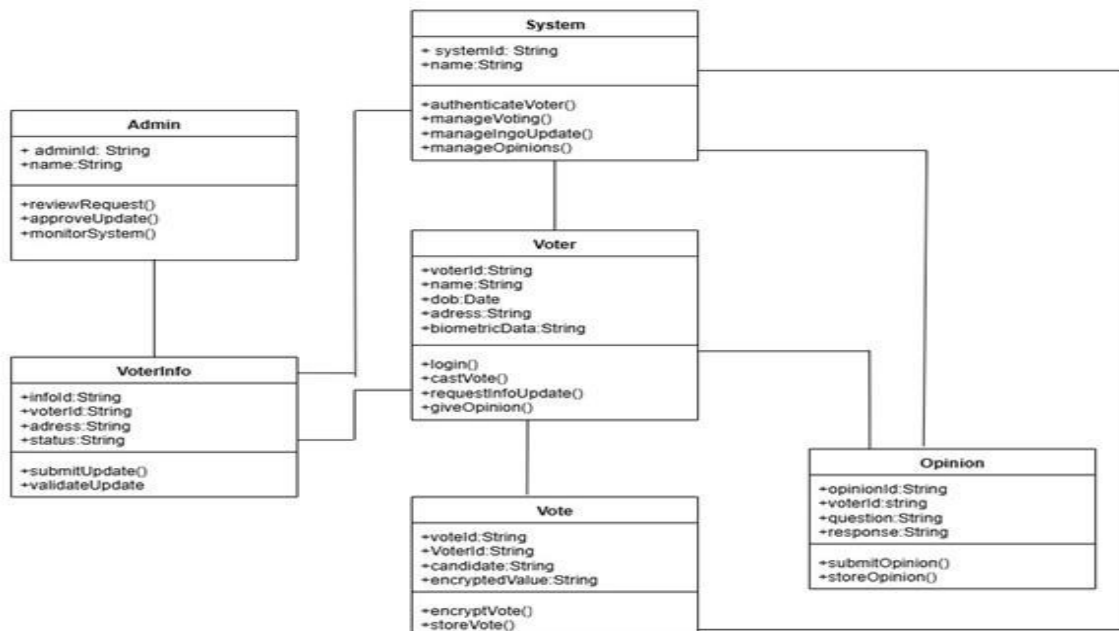
4. SOFTWARE DESIGN

4.1 System Design

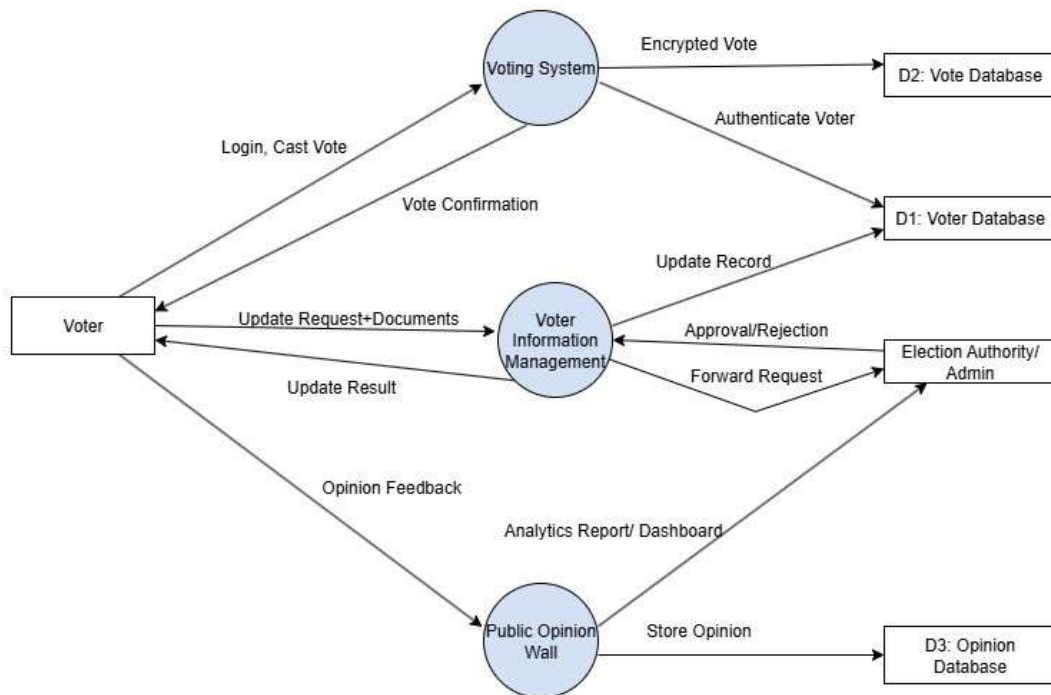
Activity diagram:



Class diagram:



DFD (data flow diagram):



4.2 UI / Wireframe Design

Home page:

The Home page wireframe features a green header with the Bangladesh Election Commission logo and a navigation menu: Home, Voter Registration, Political Parties, Voting Result, News, Public Opinion Wall, and Edit Information. A 'Login' button is in the top right. The main content area is divided into three sections: 'Important Information' with links to upcoming elections (National, City Corporation, Municipality, Union Parishad), NID Collection, Diaspora Voting, Updated Voter List, Pooling Center, and Voter Serial Collection; 'Important Links' with URLs for mygov.bd, bangladesh.gov.bd, and army.mil.bd; and 'Anonymous Opinion' polls. The first poll asks about support for a new defense research laboratory, with results: Yes (47%), No (30%), Unsure (23%), and Others. The second poll asks about support for increased government funding for aerospace innovation, with results: Yes (55%), No (30%), Unsure (15%), and Others. The third poll asks about satisfaction with the government's handling of national security issues, with results: Yes (5%), No (85%), Unsure (10%), and Others. A 'View More' button is at the bottom right. A 'Terms of Use' section is at the bottom, followed by a copyright notice: 'All Rights Reserved © 2000 - 2025. Bangladesh Election Commission'.

Login /Registration page:

The Login /Registration page wireframe features a green header with the Bangladesh Election Commission logo and the text 'Bangladesh Election Commission'. The main content area is divided into two sections: 'Don't have an account?' and 'Login'. The 'Don't have an account?' section includes a text input field for 'NID or Username', a password input field, a 'Register your account' button, and a 'Forgot Password' link. The 'Login' section includes a text input field for 'NID or Username', a password input field, a 'Login' button, and a 'Forgot Password' link. A 'Terms of Use' section is at the bottom, followed by a copyright notice: 'All Rights Reserved © 2000 - 2025. Bangladesh Election Commission'.

Public Opinion page:

The Public Opinion Wall page wireframe features a green header with the Bangladesh Election Commission logo and the text 'Welcome To Public Opinion Wall'. The main content area is divided into three sections: 'Question Category' with links to Governance and Politics, Health and Healthcare, Education, Employment and Economy, Security and Law Enforcement, Technology and Media, Military & Defence, and Other; 'Important Links' with URLs for mygov.bd, bangladesh.gov.bd, and army.mil.bd; and 'Give Your Valuable Opinion' polls. The first poll asks about support for a new defense research laboratory, with results: Yes (47%), No (30%), Unsure (23%), and Others. The second poll asks about support for increased government funding for aerospace innovation, with results: Yes (55%), No (30%), Unsure (15%), and Others. A 'View More' button is at the bottom right. A 'Terms of Use' section is at the bottom, followed by a copyright notice: 'All Rights Reserved © 2000 - 2025. Bangladesh Election Commission'.

5. GIT WORKFLOW

Git Lab Manual

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1

General

Owner *

 susmita-mitra

Repository name *

project-simulator-smart-integrated-voting

✔ project-simulator-smart-integrated-voting is available.

Great repository names are short and memorable. How about **curly-winner**?

Description

A simulation repository for practicing Git workflow management.

65 / 350 characters

2

Configuration

Choose visibility *

Choose who can see and commit to this repository

 Public

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

Off ☐

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

No .gitignore

Add license

Licenses explain how others can use your code. [About licenses](#)

No license

Create repository

```
Microsoft Windows [Version 10.0.26100.6584]
(c) Microsoft Corporation. All rights reserved.

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>git config user.name
Sudipto Kumar Chakrabarty

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>git config user.email
skchakrabarty2020@gmail.com

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/mingw64/etc/ssl/certs/ca-bundle.crt
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
```

```
Microsoft Windows [Version 10.0.26100.6584]
(c) Microsoft Corporation. All rights reserved.

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1>git clone "https://github.com/susmita-mitra/project-simulator-smart-integrated-voting.git"
Cloning into 'project-simulator-smart-integrated-voting'...
remote: Enumerating objects: 10, done.
remote: Counting objects: 100% (10/10), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 10 (delta 1), reused 8 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (10/10), done.
Resolving deltas: 100% (1/1), done.

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1>
```

```
Microsoft Windows [Version 10.0.26100.6584]
(c) Microsoft Corporation. All rights reserved.

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>git config user.name "
Sudipto Kumar Chakrabarty"

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>git config user.email"
skchakrabarty2020@gmail.com"
```

```

Microsoft Windows [Version 10.0.26100.6584]
(c) Microsoft Corporation. All rights reserved.

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>git pull origin main
From https://github.com/susmita-mitra/project-simulator-smart-integrated-voting
 * branch      main       -> FETCH_HEAD
Already up to date.

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>git checkout dev
Switched to branch 'dev'
Your branch is up to date with 'origin/dev'.

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>git pull origin dev
From https://github.com/susmita-mitra/project-simulator-smart-integrated-voting
 * branch      dev        -> FETCH_HEAD
Already up to date.

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>git checkout stage
Switched to branch 'stage'
Your branch is up to date with 'origin/stage'.

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>git pull origin stage

```

```

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>git checkout dev
Already on 'dev'
Your branch is up to date with 'origin/dev'.

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>git pull origin dev
From https://github.com/susmita-mitra/project-simulator-smart-integrated-voting
 * branch      dev        -> FETCH_HEAD
Already up to date.

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>git checkout -b feature/T-14
Switched to a new branch 'feature/T-14'

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>

```

```

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>echo "## T-14: Implement User Login Page" >> PROJECT_FEATURES.md

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>echo "This task involves creating a login form component with email and password fields, including validation and error handling." >> PROJECT_FEATURES.md

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>echo "***Status: Implemented**" >> PROJECT_FEATURES.md

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>echo "" >> PROJECT_FEATURES.md

```

```

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>git push -u origin feature/T-14
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 596 bytes | 596.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/T-14' on GitHub by visiting:
remote:   https://github.com/SUDIPTO2023/project-simulator-smart-integrated-voting/pull/new/feature/T-14
remote:
To https://github.com/SUDIPTO2023/project-simulator-smart-integrated-voting.git
 * [new branch]      feature/T-14 -> feature/T-14
branch 'feature/T-14' set up to track 'origin/feature/T-14'.

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>

```

base repository: susmita-mitra/project-simulato... base: main head repository: SUDIPTO2023/project-simulato... compare: feature/T-14

 Add a title

feat: implement user login page (T-14) Feature/t 14

Helpful resources
[GitHub Community Guidel](#)

Add a description

Write Preview H B I i <> @ ↩ ↪

feat: implement user login page (T-14) - Created login form component with fields - Added input validation - Implemented error handling for failed login attempts"



 Markdown is supported  Paste, drop, or click to add files

☒ Allow edits by maintainers  Create pull request

```
D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>git checkout dev
Already on 'dev'
Your branch is up to date with 'origin/dev'.

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>git branch -d feature/T-14
warning: deleting branch 'feature/T-14' that has been merged to
'refs/remotes/origin/feature/T-14', but not yet merged to HEAD
Deleted branch feature/T-14 (was 3fa2347).

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>
```

Git Lab Task

```
D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>git pull origin main
From https://github.com/SUDIPTO2023/project-simulator-smart-integrated-voting
* branch      main      -> FETCH_HEAD
Already up to date.
```

```

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_5
0622_1\project-simulator-smart-integrated-voting>git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_5
0622_1\project-simulator-smart-integrated-voting>git pull origin main
From https://github.com/SUDIPTO2023/project-simulator-smart-integrated-voting
 * branch          main          -> FETCH_HEAD
Already up to date.

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_5
0622_1\project-simulator-smart-integrated-voting>git checkout -b hotfix/critical-read
me-typo
Switched to a new branch 'hotfix/critical-readme-typo'

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_5
0622_1\project-simulator-smart-integrated-voting>

```

```

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_5
0622_1\project-simulator-smart-integrated-voting>git add README.md

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_5
0622_1\project-simulator-smart-integrated-voting>git add .

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_5
0622_1\project-simulator-smart-integrated-voting>git commit -m "fix: correct critical
typo in README - Changed 'implimentation' to 'implementation' - This was causing con
fusion for users"
[hotfix/critical-readme-typo e0b8677] fix: correct critical typo in README - Changed
'implimentation' to 'implementation' - This was causing confusion for users
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 git

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_5
0622_1\project-simulator-smart-integrated-voting>

```

```

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\p
roject-simulator-smart-integrated-voting>git push -u origin hotfix/critical-readme-typo
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 356 bytes | 356.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'hotfix/critical-readme-typo' on GitHub by visiting:
remote:   https://github.com/SUDIPTO2023/project-simulator-smart-integrated-voting/pull/ne
w/hotfix/critical-readme-typo
remote:
To https://github.com/SUDIPTO2023/project-simulator-smart-integrated-voting.git
 * [new branch]      hotfix/critical-readme-typo -> hotfix/critical-readme-typo
branch 'hotfix/critical-readme-typo' set up to track 'origin/hotfix/critical-readme-typo'.

```

```
D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>git pull origin main
From https://github.com/SUDIPTO2023/project-simulator-smart-integrated-voting
 * branch      main      -> FETCH_HEAD
Already up to date.

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>git checkout dev
Switched to branch 'dev'
Your branch is up to date with 'origin/dev'.

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>git pull origin dev
From https://github.com/SUDIPTO2023/project-simulator-smart-integrated-voting
 * branch      dev       -> FETCH_HEAD
Already up to date.

D:\Varsity semester\summer 25\Software Engineering\final\Smart-Integrated-Voting-23_50622_1\project-simulator-smart-integrated-voting>git branch -d hotfix/critical-readme-typo
warning: deleting branch 'hotfix/critical-readme-typo' that has been merged to
        'refs/remotes/origin/hotfix/critical-readme-typo', but not yet merged to HEAD
Deleted branch hotfix/critical-readme-typo (was e0b8677).
```

6. TESTING

Project Name: Smart Integrated Voting System with Voter Information Management & Public Opinion Wall			Test Designed by: Susmita Mitra	
Test Case ID: TC_01			Test Designed date: 26/7/2025	
Test Priority (Low, Medium, High): High			Test Executed by: Susmita Mitra	
Module Name: Login & Authentication			Test Execution date: 27/07/2025	
Test Title: Verify login with valid voter ID and OTP				
Description: Verify that the user can successfully log in using a valid voter ID and OTP.				
Precondition: The voter is already registered in the system and has access to a valid mobile number or email to receive OTP.				
Dependencies: SMS gateway and OTP service must be active				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open the login page	N/A	Home page loads within 2 seconds without any errors.	As expected	Pass
2. Enter valid Voter ID	23056231	Field accepts input	As expected	Pass
3. Enter valid OTP	543210	OTP verified successfully	As expected	Pass
4. Click the “Login” button	N/A	User is redirected to Dashboard	As expected	Pass

Project Name: Smart Integrated Voting System with Voter Information Management & Public Opinion Wall			Test Designed by: Sudipto Kumar Chakrabarty	
Test Case ID: TC_02			Test Designed date: 26/7/2025	
Test Priority (Low, Medium, High): High			Test Executed by: Sudipto Kumar Chakrabarty	
Module Name: Online Voting			Test Execution date: 29/07/2025	
Test Title: Verify that a user can cast a vote only once per election				
Description: Ensure that a registered voter cannot vote more than once in the same election.				
Precondition: The user is logged in and the election is active.				
Dependencies: The Election database and server are running.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Navigate to the Election page	N/A	Election list displays successfully	As expected	Pass
2. Select a candidate and click “Vote”	Candidate ID: 102	Vote submitted successfully	As expected	Pass
3. Attempt to vote again in the same election	Candidate ID: 103	The system displays “You have already voted” message	As expected	Pass

Project Name: Smart Integrated Voting System with Voter Information Management & Public Opinion Wall			Test Designed by: Arnob Debnath	
Test Case ID: TC_03			Test Designed date: 26/7/2025	
Test Priority (Low, Medium, High): High			Test Executed by: Arnob Debnath	
Module Name: Public Opinion Wall			Test Execution date: 28/07/2025	
Test Title: Verify that a user can submit an opinion on a government poll				
Description: Verify that a logged-in user can participate in an active poll and submit an opinion.				
Precondition: User is logged in and at least one active poll is available.				
Dependencies: Opinion Wall database must be active.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the “Opinion Wall” page	N/A	List of active polls displayed	As expected	Pass
2. Select a poll	Poll ID: 302	A question appears with options	As expected	Pass
3. Choose an option and submit	Option: “Yes”	Opinion submitted successfully	As expected	Pass
4. Refresh the page	N/A	Updated results appear on the dashboard	As expected	Pass

7. SOFTWARE PRODUCT METRICS

To evaluate the quality, performance, and maintainability of the *Smart Integrated Voting System*, several software metrics are applied:

1. Function-Based Metrics:

- Total Function Points ≈ 200 FP (based on 10,000 SLOC).
- Ensures correct functional size and scope.

2. Object-Oriented Metrics:

- $WMC \leq 15$, $DIT \leq 4$, $CBO \leq 6$, $LCOM < 0.8$.
- Measures class complexity, coupling, and cohesion for better design quality.

3. Operation-Oriented Metrics:

- Average cyclomatic complexity ≤ 10 and ≤ 4 parameters per function.
- Keeps code simple and testable.

4. Performance Metrics:

- System response time ≤ 2 seconds, uptime $\geq 99.99\%$.
- Ensures scalability and reliability for 200M users.

5. Maintenance Metrics:

- $SMI \geq 0.8$, Maintainability Index ≥ 75 .
- Tracks system stability and ease of future updates.

8. CONCLUSION AND FUTURE WORK

The Smart Integrated Voting System with Voter Information Management & Public Opinion Wall successfully demonstrates a secure, transparent, and user-friendly digital platform for voting and public engagement. This system solves major issues of traditional voting, such as accessibility, time consumption, and lack of transparency. It ensures data security through OTP, biometric verification, and encryption, while also allowing citizens to update voter information online and share opinions on government decisions. Overall, the project contributes to a more inclusive, efficient, and trustworthy democratic process.

In the future, this system can be enhanced by integrating Blockchain technology to further strengthen vote integrity and transparency. AI-based fraud detection and real-time analytics dashboards can also be implemented to improve accuracy and government responsiveness. With these advancements, the system could be adopted nationally to modernize and digitize the entire election and public feedback process.